Yiyang Wang

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EDUCATION

University of Michigan, Ann Arbor Ann Arbor, MI
Ph.D. in **Civil Engineering** (GPA: 3.96/4.00) Dec 2022
w/ specialization in *Next Generation Transportation Systems*
M.S. in **Electrical Engineering and Computer Science** (GPA: 3.81/4.00) Apr 2018
w/ specialization in *Signal & Image Processing and Machine Learning*

Jilin University Changchun, China
B.Eng. in **Communications Engineering** (GPA:90.32/100, Rank: Top 1/91) June 2016
w/ *National Scholarship Award*

SKILLS

- **Core Expertise:** Deep Learning, Reinforcement Learning, Causal Discovery/Inference, Time Series & Anomaly Detection, Combinatorial Optimization, Graph Algorithms
- **Programming Languages:** Python (Proficient), MATLAB (Proficient), SQL, R, C/C++
- **Packages & Tools:** PyTorch, Gurobi, NumPy, Pandas, Scikit-learn, GCP, TensorFlow, Git, Bash, Docker
- **Research Interests:** Machine Learning, NLP, Anomaly Detection, Multi-Armed Bandits, Combinatorial Optimization

PROFESSIONAL EXPERIENCE

Amazon | Seller Partner Service Seattle, WA
Senior Applied Scientist Dec 2025 - Present

- Lead applied scientist for a production **multimodal vision-language model (VLM)** detecting trademark/copyright infringement on Amazon storefront. Drove backbone migration (**Pixtral-12B → Qwen3-VL-8B** at 2048×2048) and designed a two-phase **SFT curriculum** on a **900K+**-sample multi-source dataset spanning **52K brand IDs** (**4.2**× prior coverage); achieved **33%** reduction of genuine false positives on shadow-launch audit with transparent recall trade-off.
- Built end-to-end **SFT + GRPO** reinforcement-learning training pipeline (EasyR1) with multi-component reward (brand-ID correctness, IP-type alignment, decision calibration); diagnosed and fixed reward-induced distributional collapse via joint-distribution checks. Multi-node training on **88×A100-40GB** HyperPod cluster (DeepSpeed ZeRO-2, two-step tokenized-data pattern on FSX/Lustre).
- Designed and executed a **353K-ASIN** auditor-aware backfill through **AWS Bedrock** (Claude Opus) to generate structured reasoning traces; integrated as supervised training data, increasing auditor-verified true-negative coverage by **20**× and lifting recall on hard true-positive subset by **+28pp** at near-flat precision.
- Trained an **MLP confidence head** on VLM hidden states for production risk scoring; deployed via **SageMaker** endpoints for shadow-audit integration. Cross-team collaboration to acquire heterogeneous data sources and address brand long-tail and compatibility-term false-positive imbalance.

Intel Corporation | Technology Development Hillsboro, OR
Machine Learning Engineer Jan 2023 - Dec 2025

- Developed novel graph-based casual discovery algorithms to unveil hidden causality among high-dimensional features with extreme limited and noisy data, to identify root sources of semiconductor variation. Close collaboration with internal customers to design real-silicon experiments, and decrease variation by at least 50% ahead of schedule and enable faster PDK target achievement. Leaded this project with a focus team, which has been identified as one of three esteemed top jobs within SVP organization.
- Built and trained supervised auto-encoder and transformer models to predict semiconductor performance using Intel inline optical spectral data and achieved best-in-class prediction performance on internal real-silicon data. The pre-trained models are launched in production for internal integration team to monitor and trend real product performance. Enabled fast and accurate performance prediction to timely identify any potential production ”excursion” (i.e. deviation from target) to prevent wafer loss.

SiriusXM & Pandora | Science Pandora Department Oakland, CA
Science Intern - Recommendation, Search, & Voice May 2022 - Aug 2022

- Built a **Siamese neural network with attention fusion** (PyTorch) for **semantic retrieval** of music on **GCP**; improved **recall by 22%** vs. production baseline and increased robustness to paraphrased/natural language queries.
- **Advanced NLP/retrieval:** engineered contrastive objectives (triplet/in-batch negatives) and indexed vectors with ANN for low-latency search at catalog scale.
- Built **PySpark** pipelines for data acquisition and query understanding (entity normalization, synonyms, spelling variants); developed offline evaluation harnesses (recall@K) with **Gensim** and **NLTK**.

Univ. of Michigan | Next Generation Mobility Systems Lab Ann Arbor, MI
Research Associate Sep 2018 - Dec 2018

- Designed an anomaly detection approach with time series trajectory data by combining **convolutional neural network (CNN)** and **Kalman filter** with χ^2-detector in **Python (PyTorch) & MATLAB** with F1 score **97.8**%
- Pre-processed the large-scale (more than 1GB) raw dataset (Safety Pilot Dataset) for training and testing using **SQL** to filter specific vehicle trajectories
- **Sensor fusion** with CNN to further improve detection performance (**14%** above benchmark) on time series dataset

Ford Motor Company | Research and Advanced Engineering (R&A) Dearborn, MI
Product Development Intern May 2018 - Jul 2018

- Forecasted the travel demand in 5 and 10 years of Ann Arbor city using a **four-step travel demand model**
- Used **logistic regression** for travel mode choice prediction, and **gravity model** for trip distribution prediction
- Predicted and visualized the traffic congestion level on each road in Ann Arbor city with **SUMO**, specified the roads need expansion

China Unicom | Network Management Center Jinan, China
Network Telecommunications Engineer Intern Jul 2015 - Sep 2015

- Enabled rapid and dynamic IP assignment to all China Unicom internet customers in Jinan city, by pre-allocating IP address resources in the IP address resources management system
- Tested the packet loss rate with **secureCRT** and fixed the line failures

RESEARCH EXPERIENCE

Dynamic Security Resource Allocation in Connected and Automated Vehicles
Python Aug 2022 - Present

- Formulated a partially observable Markov Decision Process (POMDP) to prescribe an optimal policy for dynamical security resource allocation during the trip, which ensures the security and energy efficiency of CAVs
- Solved the POMDP model using **point based value iteration (PBVI)** algorithm in **Python**

Demand Forecasting and Vehicle Route Planning Algorithm in Benton Harbor
Python (Gurobi), MATLAB Jan 2021 - Jan 2022

- Forecasted travel demand and designed new transit routes in Benton Harbor to improve mobility for local residents, which increased the annual ridership up to **78%**
- Trained **radial basis function (RBF) network for regression**, with socioeconomic data, for travel demand forecasting by **MATLAB** with high accuracy (**RMSE 4.93**)
- Proposed and solved a demand-responsive optimization model in **Python & Gurobi** on large-scale datasets (preprocessed by **SQL**), which provided the optimal new bus routes
- Devised a graph aggregation-disaggregation algorithm (**Python** & **Gurobi**) which dynamically clustered the large-scale network to **reduce computation time**, and efficiently recovered from the aggregated solution (w/ convergence guaranteed)

Deep Reinforcement Learning-Bayesian Framework for Anomaly Detection
Python (PyTorch) July 2020 - Dec 2020

- Developed a POMDP model, which was solved by a novel and effective deep reinforcement learning algorithm (**A3C**), to online update CNN detecting anomalies in vehicle sensor data
- Outperformed state-of-the-art benchmarks (**12%** above CNN, **18%** above RNN) on large-scale dataset (Safety Pilot Dataset)

Adversarial Online Learning with Variable Plays in Sequential Game for Vehicle Cybersecurity
Python Sep 2019 - Oct 2020

- Devised a fast (no-regret) algorithm for the **adversarial multi-armed bandit with variable plays (MAB-VP)** problem to predict adversarial behaviours and tested on real dataset (Car-Hacking Dataset)
- Showed two directions on improving the cybersecurity from a game-theoretical perspective (**two-player sequential constant-sum games**): increase threat-monitoring resources, and/or increase reliability of the system

Anomaly Detection in Connected & Automated Vehicle Sensors
Python, MATLAB Jan 2019-Dec 2019

- Proposed an anomaly detection method for time series trajectory data by combining **Kalman filter** with unsupervised learning **One Class Support Vector Machine (OCSVM)** models, achieved AUC score **0.98/1.00** (**23%** above χ^2-detector benchmark)
- Conducted stability analysis of the platoon dynamics under cybersecurity uncertainties
- Derived an **augmented-state formulation** to further boost detection performance (up to **27%**) under **stochastic time delay**

TEACHING EXPERIENCE

CEE 373: Statistical Methods for Data Analysis and Uncertainty Modeling, Univ. of Michigan
Graduate Student Instructor — Sep 2020 - Dec 2020
Sep 2019 - Dec 2019

PUBLICATIONS

- **"Real-time Sensor Anomaly Detection and Identification in Automated Vehicles."** IEEE Transactions on Intelligent Transportation Systems [**Paper**] *(428 citations)*
- **"Real-Time Sensor Anomaly Detection and Recovery in Connected Automated Vehicle Sensors."** IEEE Transactions on Intelligent Transportation Systems [**Paper**] *(207 citations)*
- **"Anomaly detection in connected and automated vehicles using an augmented state formulation."** 2020 Forum on Integrated and Sustainable Transportation Systems (FISTS) [**Paper**] *(31 citations)*
- **"Adversarial Online Learning with Variable Plays in the Pursuit-Evasion Game: Theoretical Foundations and Application in Connected and Automated Vehicle Cybersecurity."** IEEE Access [**Paper**] *(7 citations)*
- **"A Dynamic Deep Reinforcement Learning-Bayesian Framework for Anomaly Detection."** IEEE Transactions on Intelligent Transportation Systems [**Paper**] *(63 citations)*
- **"Anomaly Detection and String Stability Analysis in Connected Automated Vehicular Platoons."** Transportation Research Part C [**Paper**] *(34 citations)*
- **"Improving Transit in Small Cities through Collaborative and Data-driven Scenario Planning."** Case Studies on Transport Policy [**Paper**] *(10 citations)*
- **"Cybersecurity in Connected and Automated Transportation Systems."** Ph.D. Thesis [**Paper**]
- **"Dynamic Security Resource Allocation for Connected and Automated Vehicles."** Transportation Research Part C [**Paper**]
- **"An Aggregation/Disaggregation Algorithm for Transit Route Planning Problem."** Under Review

TALKS AND PRESENTATIONS

Dynamic Resource Allocation for Connected and Automated Vehicles' Cybersecurity.
- TRB Annual Meeting, Washington DC, Jan. 2024.

Anomaly Detection and String Stability Analysis in Connected Automated Vehicular Platoons.
- INFORMS Annual Conference, Phoenix, AZ, Oct. 2023.
- TRB Annual Meeting, Washington DC, Jan. 2023.
- Mcity Research Review, Ann Arbor, MI, Nov. 2022.
- Mcity's Cyber Working Group. Oct. 2022. (virtual)
- NGTS Seminar, Department of Civil and Environmental Engineering, University of Michigan, Ann Arbor, Sept. 2022.

Real-Time Sensor Anomaly Detection and Recovery in Connected Automated Vehicle Sensors.
- Bridging Transportation Researchers (BTR) Conference. Aug. 2022. (virtual)
- International Symposium on Transportation Data and Modelling, Ann Arbor, MI. Jun. 2021. (virtual)
- International Symposium on Transportation Data and Modelling, Ann Arbor, MI. June. 2020. (postponed)
- INFORMS Annual Conference, Seattle, WA, Oct. 2019.
- IAVSD Workshop on Dynamics of Road Vehicles Connected and Automated Vehicles, Ann Arbor, MI, Apr. 2019.
- NGTS Seminar, Department of Civil and Environmental Engineering, University of Michigan, Ann Arbor, Jan. 2019.

Anomaly Detection in Connected and Automated Vehicles Using an Augmented State Formulation.
- Forum on Integrated and Sustainable Transportation Systems (FISTS), Nov. 2020. (virtual)

Adversarial Online Learning with Variable Plays in the Evasion-and-Pursuit Game: Theoretical Foundations and Application in Connected and Automated Vehicle Cybersecurity.
- NGTS Seminar, Department of Civil and Environmental Engineering, University of Michigan, Ann Arbor, Oct. 2020. (virtual)

A Data-Driven Framework for Optimizing Transit Itineraries.
- 2019 Michigan Institute for Data Science Symposium, Ann Arbor, MI, Nov. 2019.

ACADEMIC SERVICES

Conference Reviewer:
- International Symposium on Transportation and Traffic Theory (ISTTT): 2025
- Transportation Research Board Annual Meeting (TRBAM): 2020 - 2024
- Bridging Transport Researchers Conference (BTR): BTR4
- IEEE International Conference on Intelligent Transportation Systems (ITSC): 2021 - 2020

Journal Reviewer:
- IEEE Network Magazine
- IEEE Sensors Journal (IEEE Sens. J.)
- IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)
- IEEE Transactions on Vehicular Technology (IEEE TVT)
- Peer-to-Peer Networking and Applications
- Space: Science & Technology
- Wireless Communications and Mobile Computing

HONORS

- **William S. Housel Fellowship**, University of Michigan, Ann Arbor — Jan 2019 - Dec 2019
- **Outstanding Graduates Honer**, Jilin University — Apr 2016
- **Posts and Telecommunications Alumni Scholarship (top 2%)**, Jilin University — Sep 2015 - Apr 2016
- **Dong-Rong Scholarship (top 3%)**, Jilin University — Sep 2015 - Apr 2016
- **First Prize Scholarship (top 5%)**, Jilin University — Sep 2015 - Apr 2016
- **National Scholarship (top 1/91)**, Jilin University — Sep 2014 - Apr 2015
- **First Prize Scholarship (top 5%)**, Jilin University — Sep 2013 - Apr 2014

LEADERSHIP

Michigan Transportation Student Organization (MiTSO) | Treasurer
University of Michigan, Ann Arbor — Sep 2019 - Apr 2022

Michigan Transportation Student Organization (MiTSO) | Secretary
University of Michigan, Ann Arbor — Sep 2019 - Apr 2020